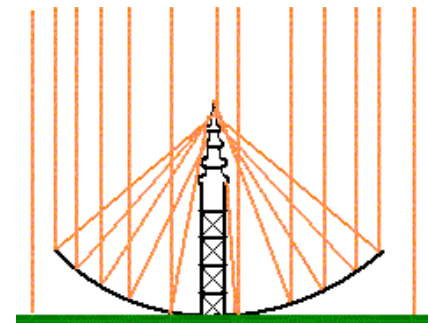
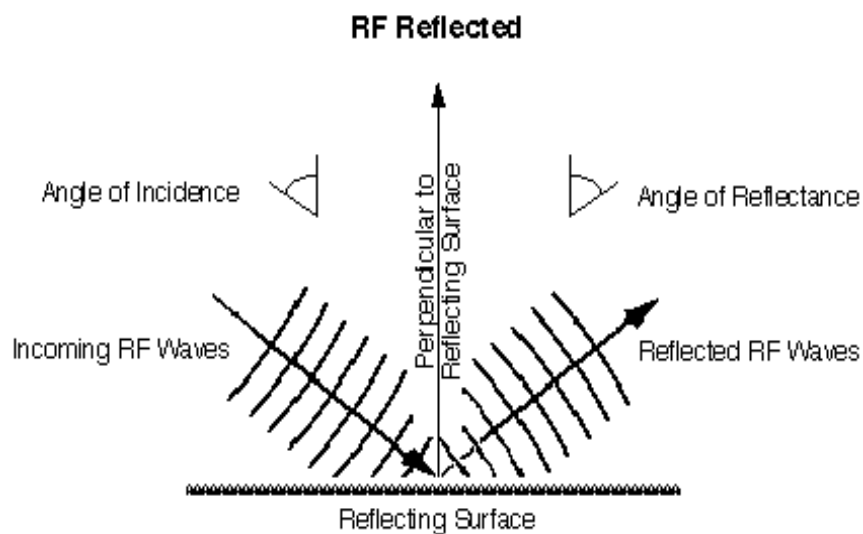


Signal Propagation

13

□ Reflection

- Propagation wave impinges on an object larger than its wavelength, e.g., buildings, walls



**Boost WiFi signals
using a beer can!**

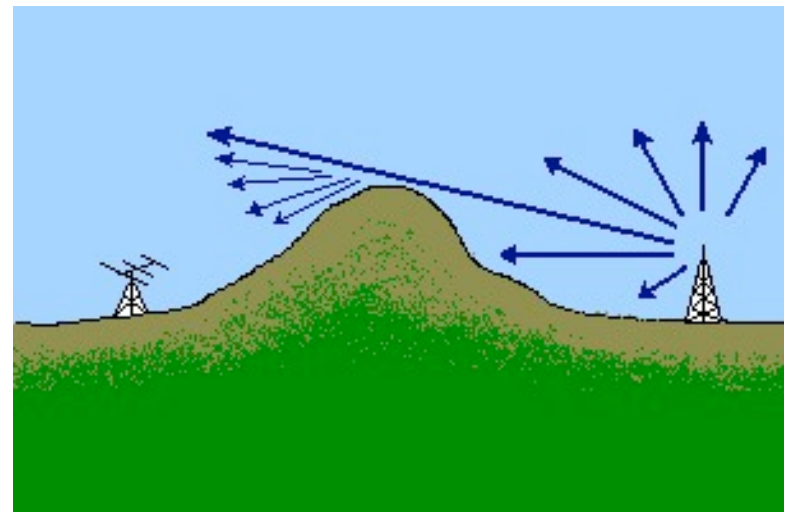
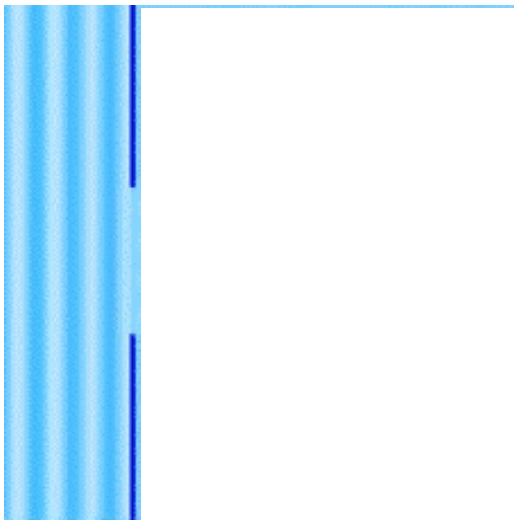


Signal Propagation

14

□ Diffraction (shadowing)

- Radio path between TX and RX are blocked by surface with sharp irregular edges
- Waves bend around the obstacle, even when line of sight (LOS) does not exist



Signal Propagation

15

□ Scattering

- Like diffraction, but at objects smaller than the wavelength of the propagation wave
- e.g., foliage, street signs, lamp posts
- The most difficult to capture



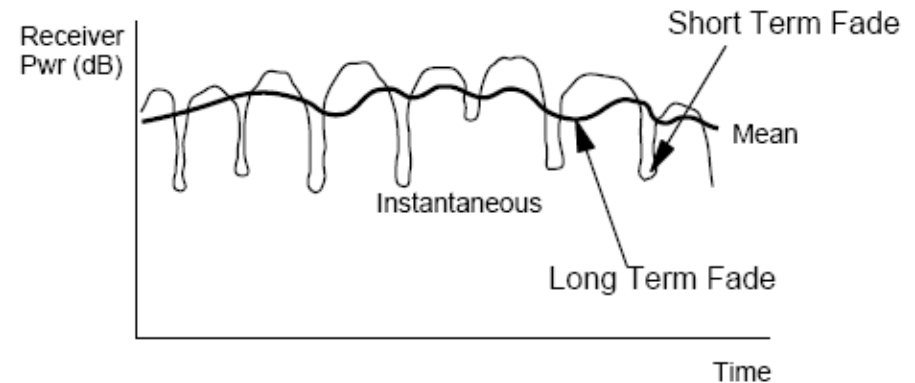
The Impact of Mobility

16

- Mobility creates **signal fluctuations**

- **Large-scale fading**

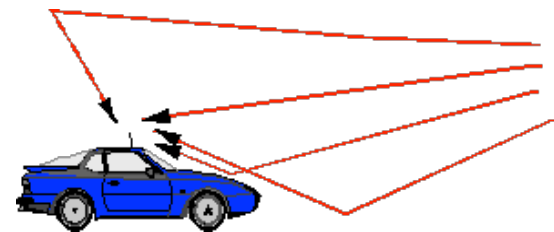
- Attenuation: in free space, power degrades by $1/d^2$
 - Shadows: signals blocked by obstacles



- **Small-scale fading**

- **Multipath effects**

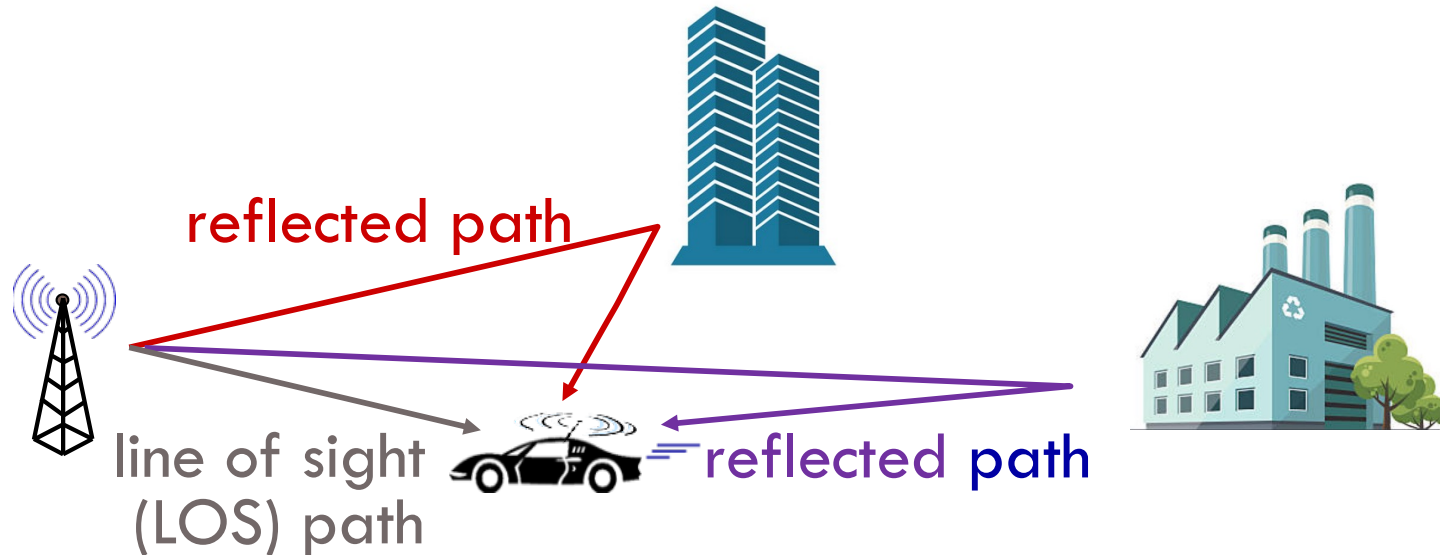
- Rapid changes in signal strength in a small area or time interval
 - Even when mobile is stationary, the received signals may fade because of the movement of surrounding objects



Multipath Fading

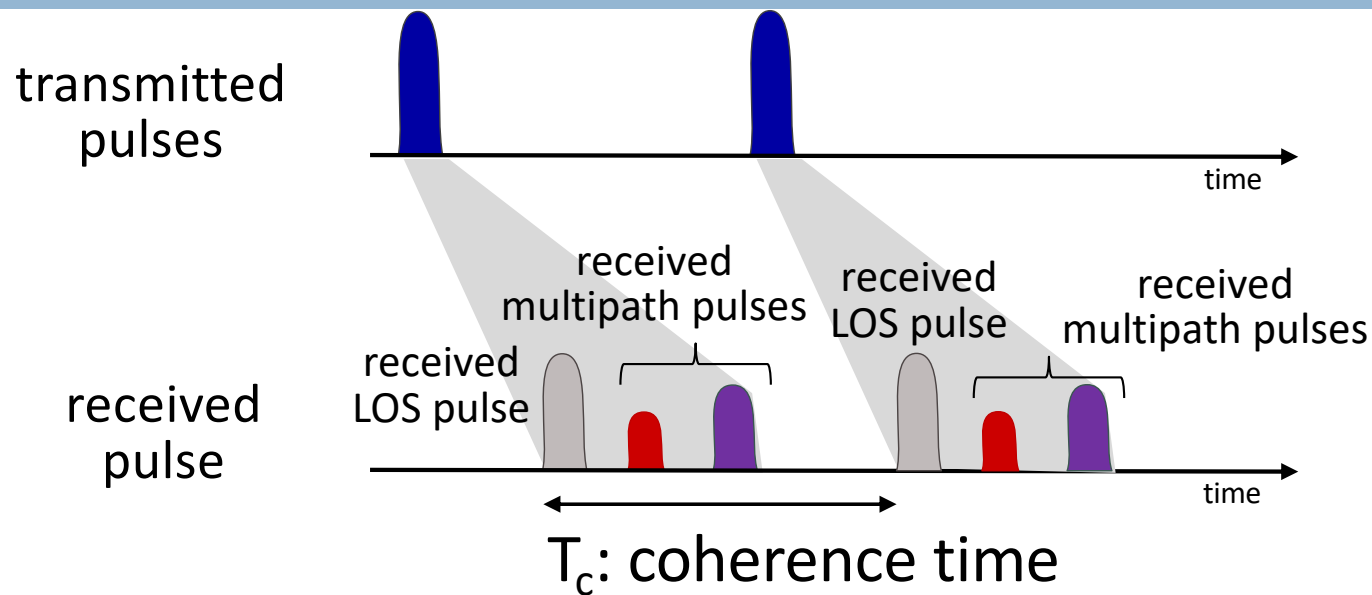
17

- Signals from different paths arrive at the receiver at different times
- Cause inter-symbol interference, limit maximum symbol rate



Multipath

18



□ Coherent time

- ▣ Time duration over which the channel impulse response is invariant
- ▣ influences maximum possible transmission rate, since coherence times can not overlap

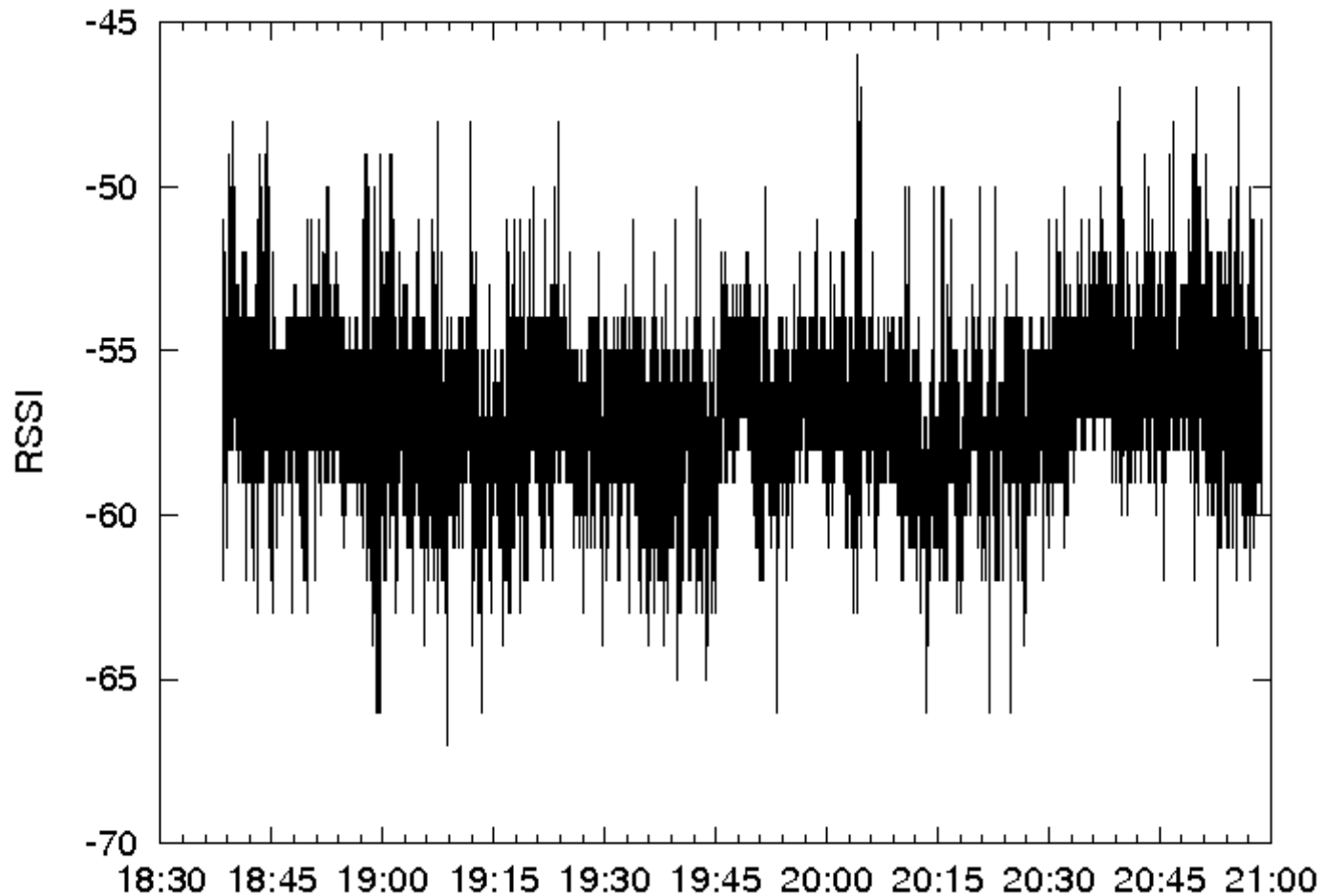
Noise and Interference

19

- Interference from other sources on the same wireless frequencies
- Two important metrics evaluate the quality of a wireless link
 - ▣ Signal to noise ratio (SNR) in dB: $10 * \log_{10} \frac{Signal}{Noise}$
 - ▣ Signal to interference and noise ratio (SINR) in dB: $10 * \log_{10} \frac{Signal}{Noise + Interference}$
- SNR/SINR vs bit error rate (BER)
 - ▣ Higher SNR/SINR → Lower BER

Received Signal Over Time

20



Challenging to model wireless propagation!

Simulating Wireless Environment

24

- A tough task
- Depends on indoor or outdoor
- Attenuation in dB

$$P_L(d) = P_L(d_0) + 10 \alpha \log(d/d_0) + X$$

 **Reference point**

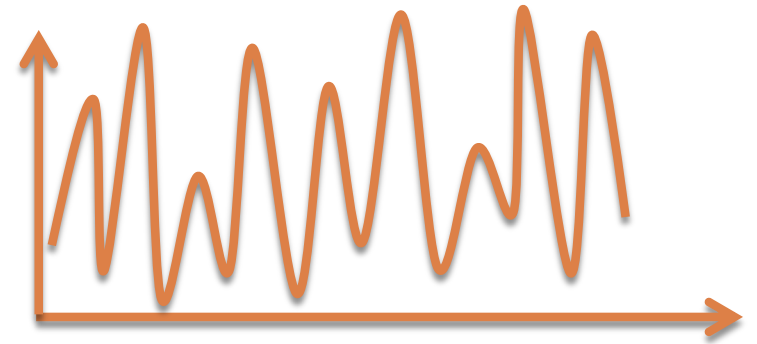
- Testbed and real measurements are highly appreciated!

Difference Between Wireless and Wired

25

- **Time-varying**, often unpredictable channel quality

- Nature of radio propagation, mobility, dynamic environment



- **Interference**

- Broadcast nature



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Wireless MAC Protocols

The More, The Messier

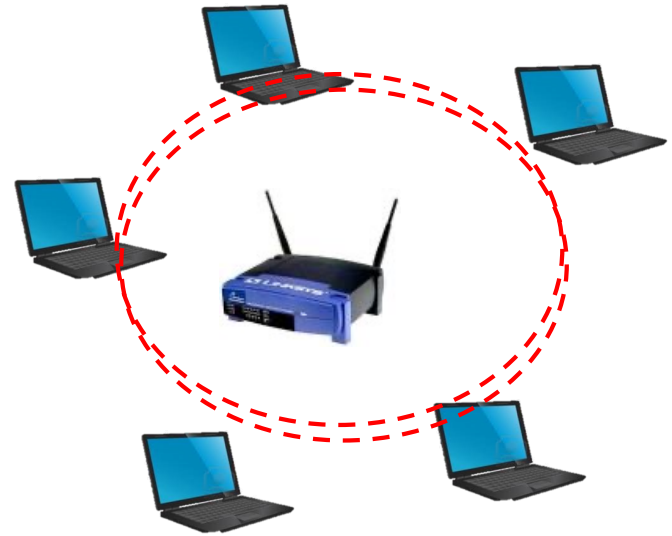
27



Role of MAC

28

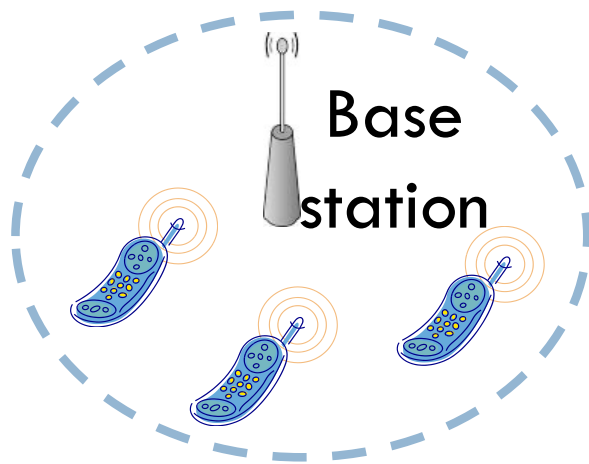
- Select transmission rate to adapt to channel variations
- Avoid interference
- Provide fairness



MAC Categories

29

Centralized

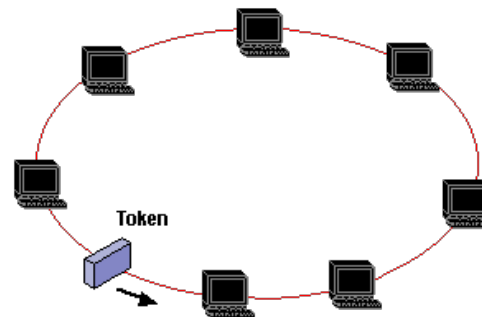


Cellular network

BS assigns time slots/frequencies/codes to users (TDMA, FDMA, CDMA)

Decentralized

Controlled access



Channel reservation (token)

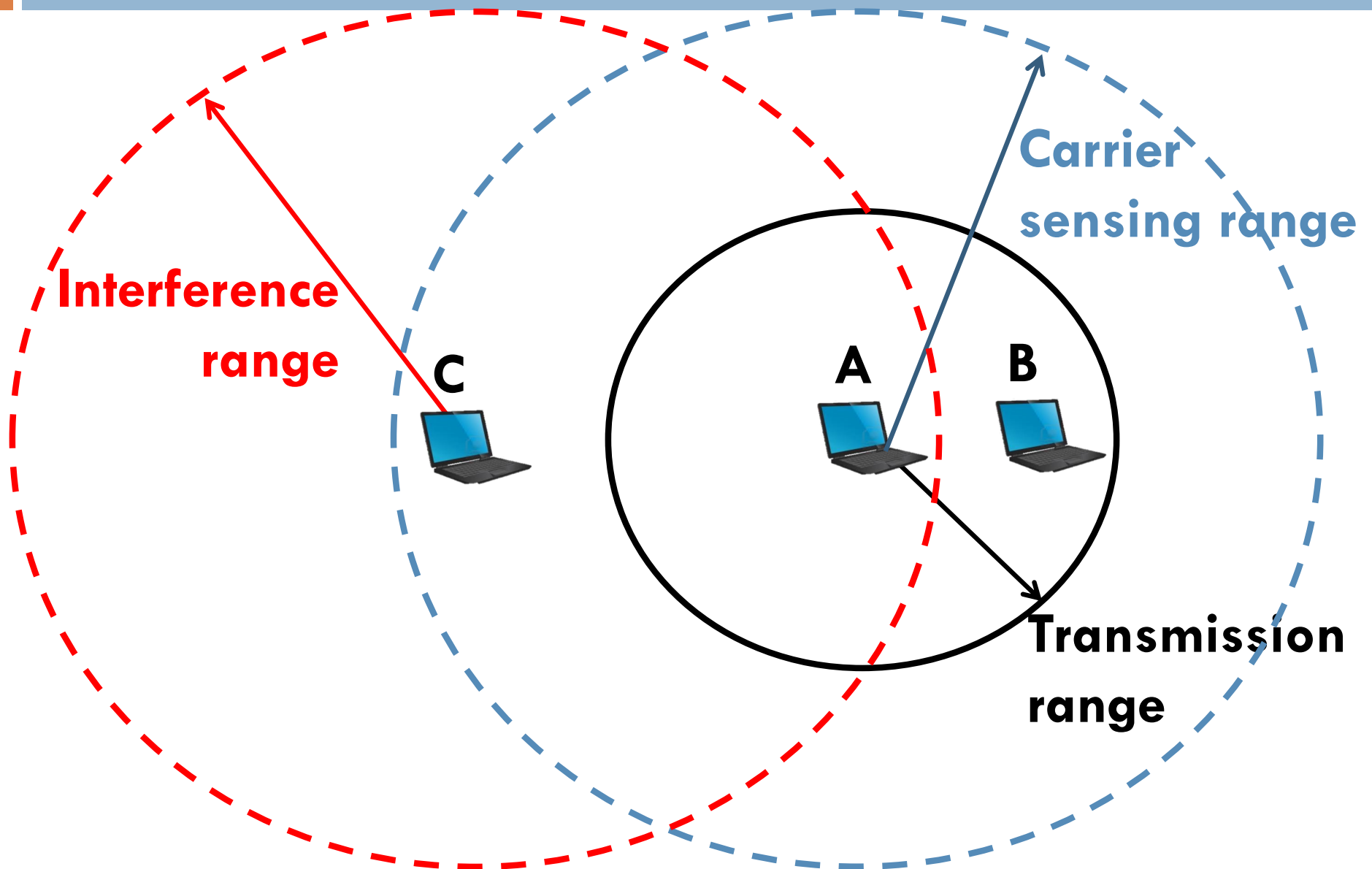
Random access



- 1) ALOHA
- 2) 802.11, CSMA/CA
- 3) TDMA, FDMA

All These Ranges

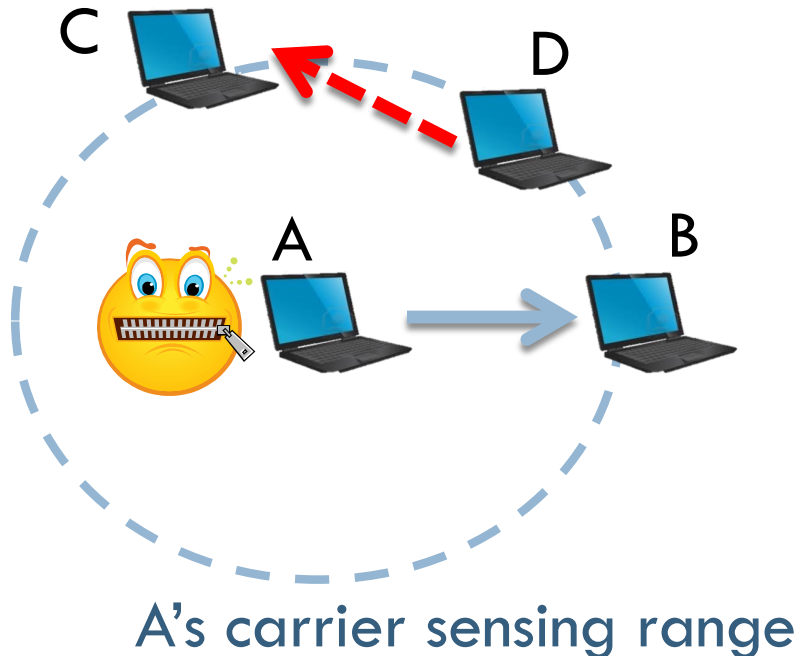
30



Carrier Sense Medium Access (CSMA)

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- Sense before transmitting
 - ▣ Channel idle: transmit
 - ▣ Channel busy: defer transmission to avoid collision



CSMA is sender-driven

Discussion Time

32

□ Is CSMA good enough?



Discussion Time

33

- Is CSMA good enough?

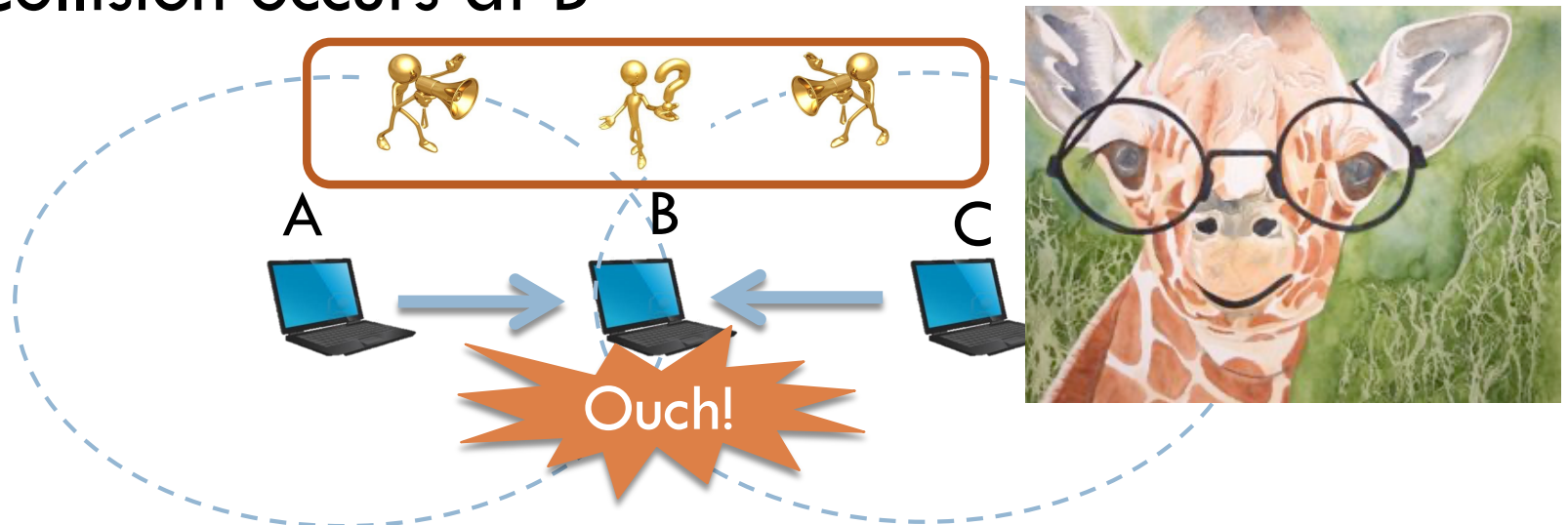


CSMA is sender-driven,
but interference is **receiver-driven**

Hidden Terminal Problem

34

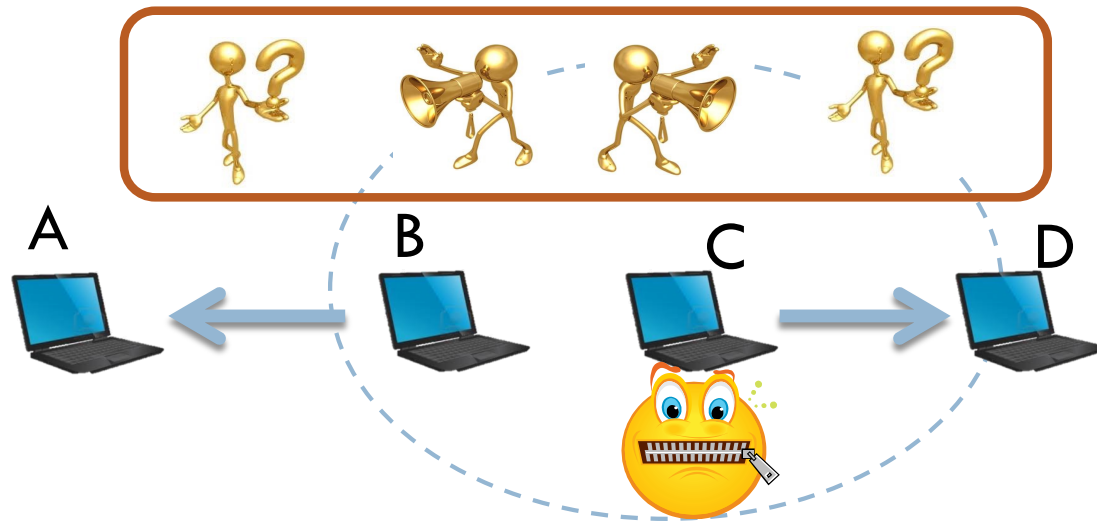
- Limited sensing range →
Cannot always sense ongoing transmissions
- A is communicating with B
- C cannot sense A's transmission, so C also transmits
→ Collision occurs at B

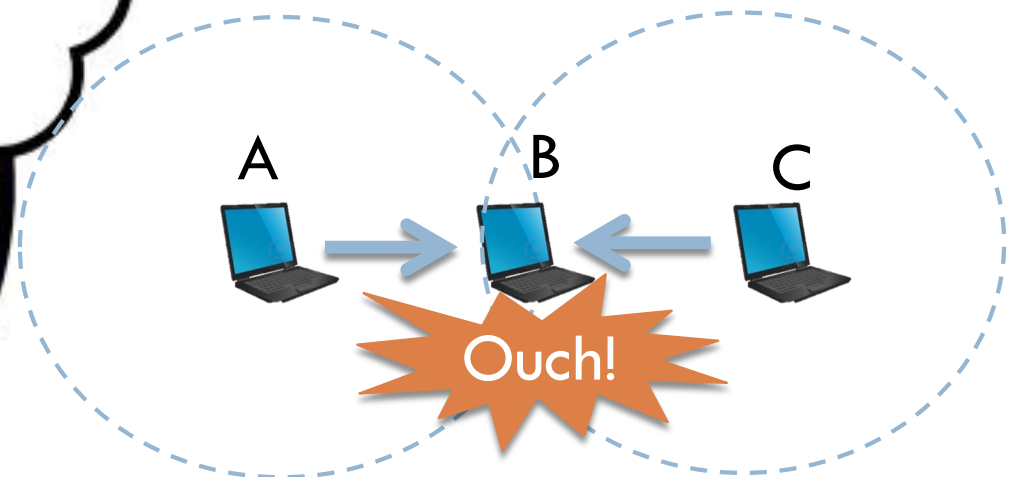
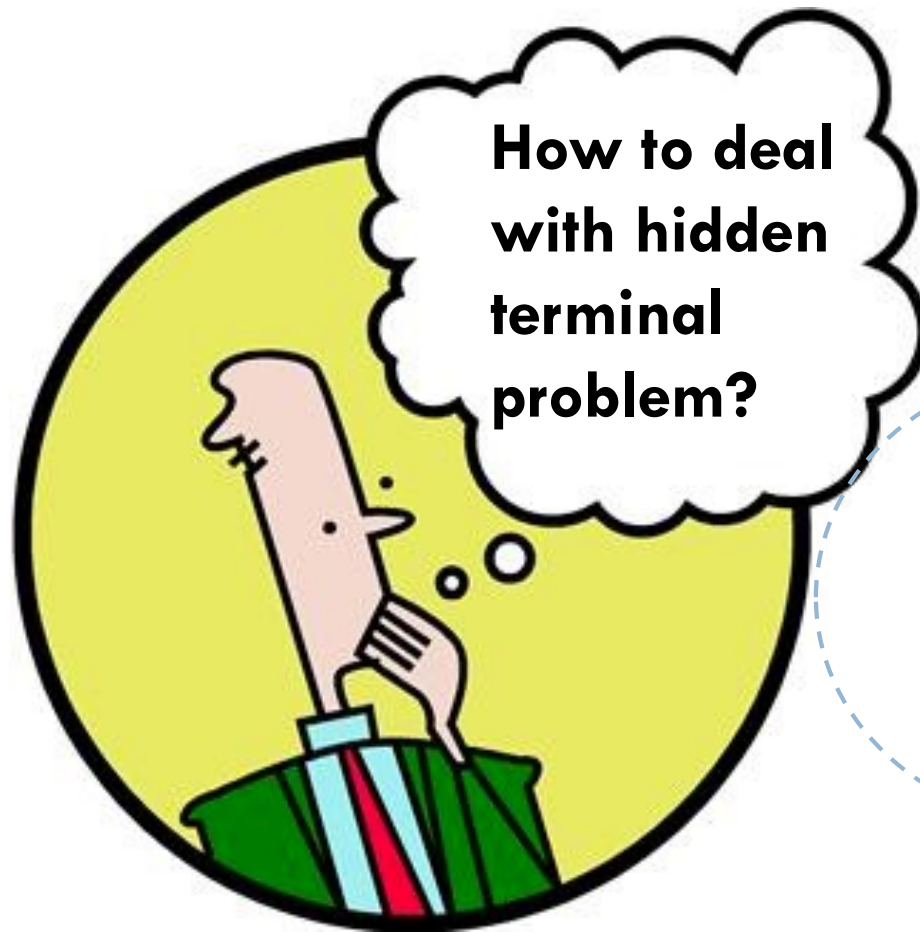


Exposed Terminal Problem

35

- Defer transmissions when it's not necessary
- C wants to transmit to D but hears B
- C defers its transmission although it won't disturb A's reception

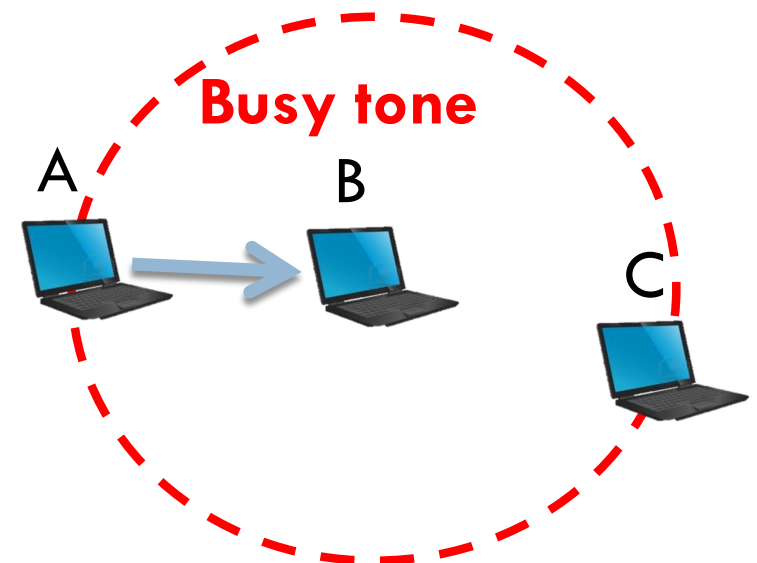
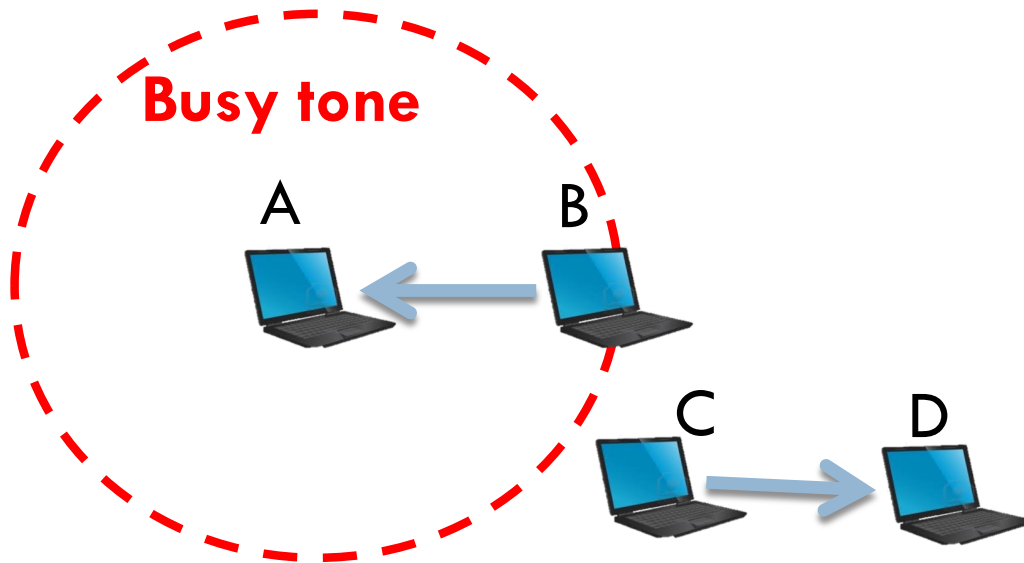




Solution #1: Avoid it

37

- Busy tone
 - **Receiver** broadcasts a busy tone while receiving data
 - Nodes hearing busy tone keep silent



Solution #1: Avoid it

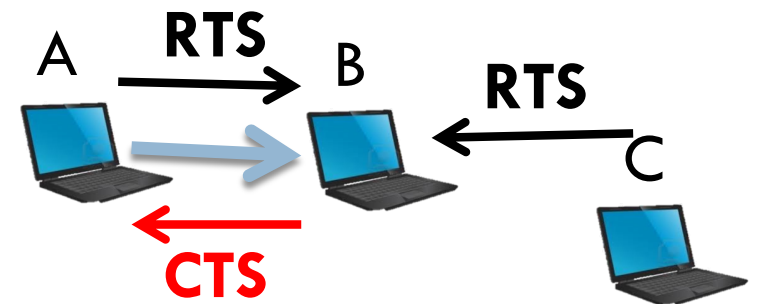
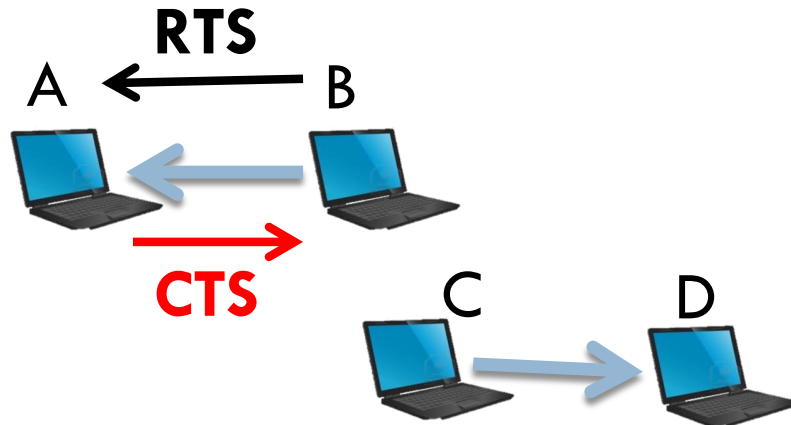
38

- MACA* (Karn90)
 - ▣ Transmitter sends a **Request-to-Send (RTS)** to receiver
 - ▣ Receiver responds with a **Clear-to-Send (CTS)**
 - ▣ Nodes hearing RTS keep silent for a short time so that transmitter can receive CTS
 - ▣ Nodes hearing CTS keep silent for the duration of the transmission

* MACA - A New Channel Access Method for Packet Radio. Philip Karn.

Example of RTS and CTS

39



802.11 MAC (CSMA/CA)

40

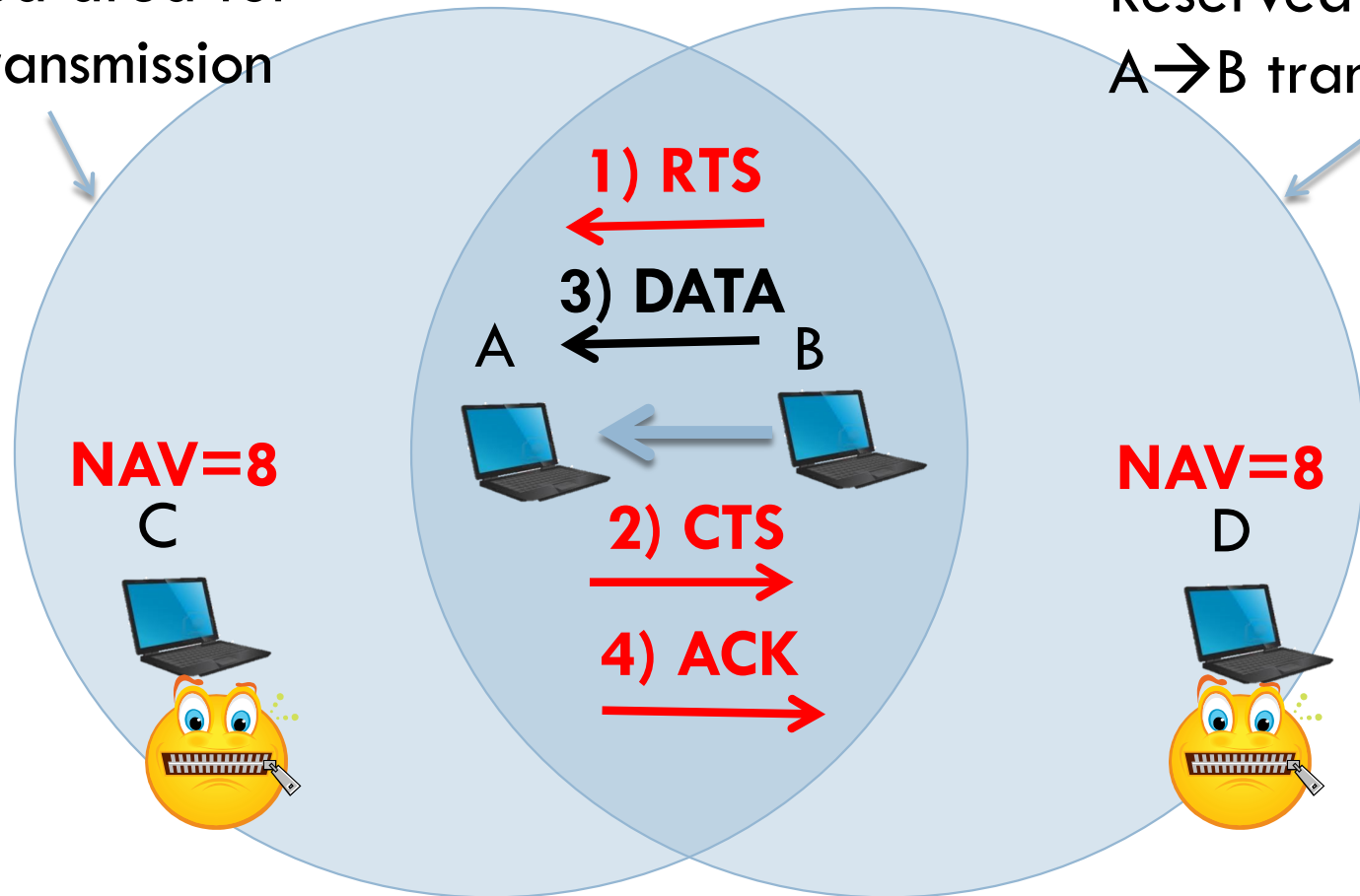
- Carrier Sense Medium Access/**Collision Avoidance**
- Carrier sense
 - ▣ When a node wants to transmit, it waits until the channel is idle
- Collision avoidance
 - ▣ RTS and CTS packets carrying **network allocation vector (NAV)** (**remaining duration to keep quiet**)
 - ▣ Nodes overhearing RTS keep silent for NAV (avoid collision w/ ACK)
 - ▣ Nodes overhearing CTS keep silent for NAV (avoid HT)
 - ▣ Once channel is idle, wait for randomly chosen duration before transmitting again → **backoff**

CSMA/CA Example

41

Reserved area for
 $B \rightarrow A$ transmission

Reserved area for
 $A \rightarrow B$ transmission



CSMA/CA

42

□ Carrier Sense Medium Access/**Collision Avoidance**

□ Carrier sense **Physical carrier sense (energy detection)**

- When a node wants to transmit, it waits until the channel is idle

□ Collision avoidance **Virtual carrier sense (packet sniffing)**

- RTS and CTS packets carrying NAV (remaining duration to keep quiet)
- Nodes overhearing CTS keep silent for NAV (avoid HT)
- Nodes overhearing RTS keep silent for NAV (avoid collision w/ ACK)

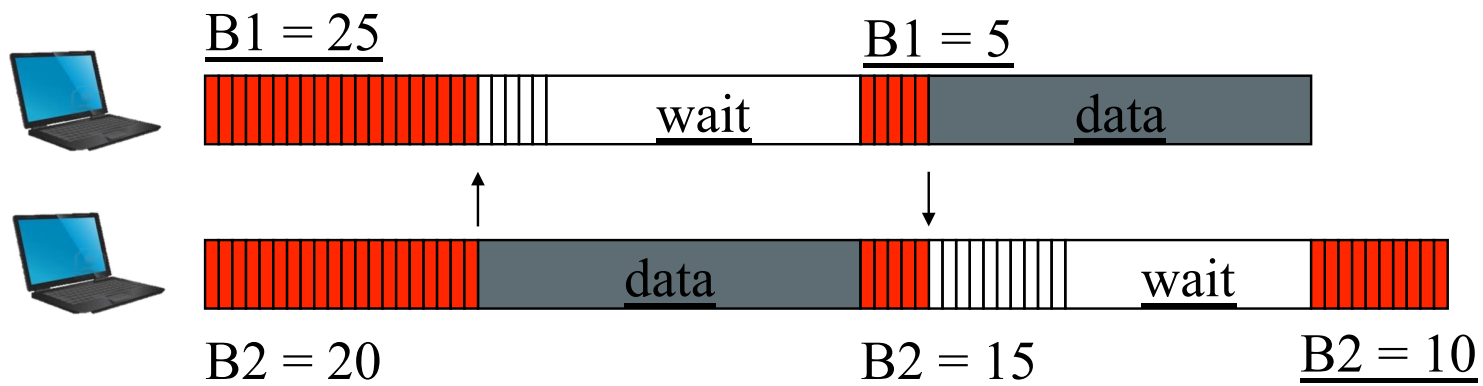
Collision avoidance (random backoff)

- Once channel is idle, wait for randomly chosen duration before transmitting again → **backoff**

Random Backoff

43

- When transmitting a packet, choose a backoff interval within $[0, CW]$
 - ▣ **CW: contention window**
- Count down the backoff interval when the medium is idle
 - ▣ Freeze the count-down if medium is busy
- When backoff interval reaches 0, transmit RTS



Put All Together: Transmit or Not Transmit?

44

- Sense the channel

→ If the channel is idle, wait for the **DIFS** interval (Distributed Inter Frame Space)

- If the channel is still idle, transmit
- For response packets (e.g., CTS, ACK), wait for **SIFS** interval (Short Inter Frame Space)

- *Why?*

→ If the channel is busy, wait until it's free

- Realize it after **DIFS** (the time needed to sense the channel)
- Start random backoff process
- Transmit when the backoff timer counts down to zero
- If no ACK, increase the backoff interval

How to Choose the Backoff Interval?

45

- The time spent on counting down backoff interval is part of MAC overhead
 - ▣ Channel is idle!

- Tricky to pick the right CW
 - ▣ Large CW → large MAC overhead
 - ▣ Small CW → many collisions

- What's your solution?

802.11 DCF

(Distributed Coordination Function)

46

- CW depends the #of contending nodes
 - ▣ Varies over time

→ Adapt CW based on collision occurrence

Binary Exponential Backoff

47

- When a node fails to receive CTS in response to RTS, it **doubles** its CW
- When a node successfully completes a data transfer, **restore CW to CW_{min}**
 - ▣ CW_{min}=31 for 802.11b, and 15 for 802.11g
- Any problems?
 - ▣ Fast oscillation in CW
 - ▣ Unfairness occurs when one node has backed off much more than other nodes

Key reason: CW is reduced too fast!

MACAW* Solution

48

- Update CW by **exponential increase, linear decrease**
 - ▣ When a node successfully transfers data, reduces CW by 1
 - ▣ *How does it compare to TCP's congestion control?*
- Append CW value to the packet
 - ▣ Each node maintains CW independently
 - ▣ CW as an indication of the level of congestion in the receiver vicinity

* MACAW: A Media Access Protocol for Wireless LAN's. SIGCOMM'94
(<http://www.cs.mcgill.ca/~cs535/docs/BhD94.pdf>)

Solution #2: Deal with it

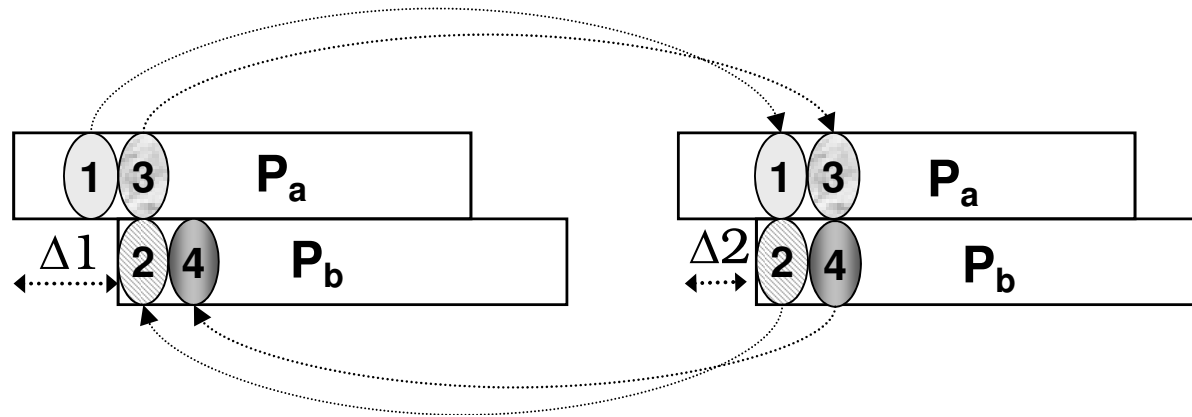
50

- Can we design receiver to cancel/utilize the impact of interference?

ZigZag Decoding

51

- Leverage two facts of 802.11 transmissions
 - ▣ Sender **retransmits** packet until it's ACKed or timed out
 - ▣ Senders **jitter** every transmissions by a random interval



Step #1: decode chunk 1 from the 1st collided packet

Step #2: subtract chunk 1 from 2nd collided packet and decode chunk 2

Step #3: subtract chunk 2 from the 1st collided packet to get chunk 3

....

Zigzag decoding: combating hidden terminals in wireless networks. SIGCOMM 2006.

IEEE 802.11 Wireless Protocols

52

- 802.11b (1997/1999)
 - ▣ 2.4 GHz, DSSS/FHSS, up to 11 Mbps
- 802.11a (1999)
 - ▣ 5 GHz, OFDM, up to 54 Mbps
- 802.11g (2003)
 - ▣ 2.4 GHz, OFDM, up to 54 Mbps
- 802.11n (2009)
 - ▣ 2.4/5 GHz, MIMO-OFDM (at most 4 MIMO streams), 300 – 500 Mbps
- 802.11ac, Wi-Fi 5 (2013)
 - ▣ 5 GHz, MIMO-OFDM (at most 8 MIMO streams), > 1 Gbps!

IEEE 802.11 Wireless Protocols

53

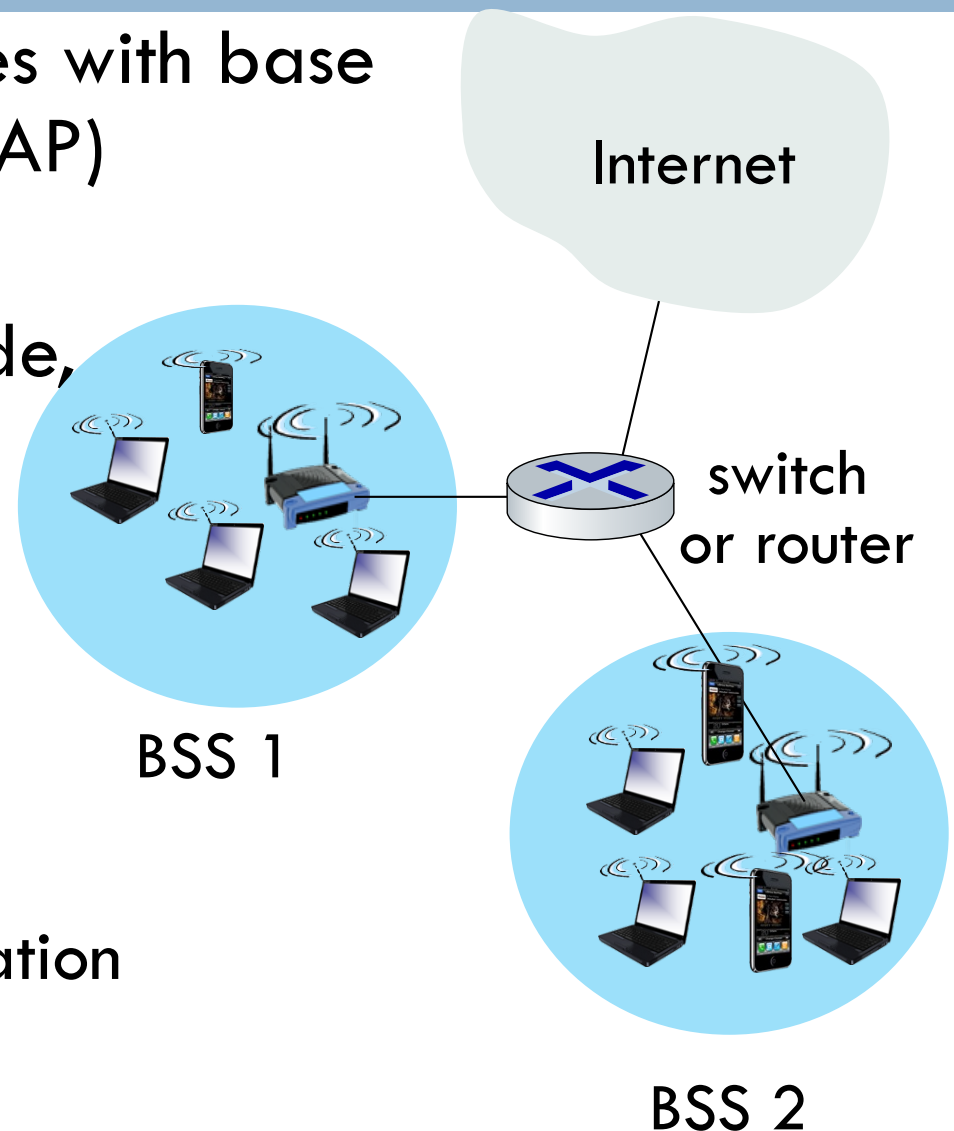
- 802.11af, White-Fi (2014)
 - ▣ Unused TV bands (54 – 590 MHz) -- TV whitespaces, 35 - 560 Mbps, 1-km range
- 802.11ah, Wi-Fi HaLow (2017)
 - ▣ 900 MHz, 347 Mbps, 1-km range, low power consumption, supporting IoT
- 802.11ax, Wi-Fi 6 (2020)
 - ▣ 2.4 GHz, 5 GHz, max 14 Gbps, better performance for crowded places

They all use CSMA/CA for medium access control

802.11 LAN Architecture

54

- wireless host communicates with base station, i.e., access point (AP)
- Modes: Infrastructure mode, ad-hoc mode
- Basic Service Set (BSS) in infrastructure mode has:
 - ▣ wireless hosts
 - ▣ access point (AP): base station
 - ▣ ad hoc mode: hosts only



802.11 Channels

55

- 802.11b: 11 channels in US
 - ▣ Each channel is 20+2 MHz wide
 - ▣ Three orthogonal channels

